

AI and Machine Learning Reference

Introduction to Machine Learning

Machine learning enables computers to learn patterns from data and use those patterns to make predictions or decisions.

Supervised Learning

Supervised learning uses labeled data. Classification predicts discrete categories, while regression predicts continuous numerical values.

Model Training

A machine learning model learns relationships from training data. The training process should be evaluated using data that was not used to train the model.

Feature Scaling

Feature scaling transforms numerical features to comparable ranges. It can improve the performance of algorithms such as support vector machines, k-nearest neighbors, and logistic regression.

Hyperparameter Tuning

Hyperparameter tuning searches for model settings that improve performance. Grid search and randomized search are common approaches.

Cross-Validation

Cross-validation divides available training data into multiple folds. It helps estimate how well a model may generalize to unseen data.

Classification Metrics

Precision measures how many predicted positive cases are actually positive. Recall measures how many actual positive cases are correctly identified. The confusion matrix summarizes classification results.

Deep Learning

Deep learning uses neural networks with multiple layers. These models can learn complex patterns from large datasets.

Natural Language Processing

Natural language processing, or NLP, enables computers to process and understand human language. NLP is used in applications such as question answering, text classification, and information retrieval.

Vector Databases

Vector databases store numerical representations called embeddings. They can retrieve documents based on semantic similarity between a query and stored document embeddings.